

BrickAtlas LDraw-2: Instance-Grounded Modular Evaluation of Visual, Source-Data, Graph and Scene-Edit Competencies

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Abstract

Assembly evaluation must distinguish recognizing a numbered part from reading its source attributes, operating on supplied connectivity, and following an editing contract. BrickAtlas LDraw-2 aligns these four observations through stable instance identity across 24 original LDraw/OMR sources, 15,334 part instances and 617 items in eleven families. Individual questions evaluate modular competencies: no family currently requires chaining two observation contracts. Standard visual items use operand-isolated images; source scale describes corpus diversity, not established per-item difficulty. We repair four deterministic generation shortcuts, separate internal answers from actual model requests, and independently check every item against source or public evidence. We provide paired full-scene, masked and cropped renders, an executable evaluation protocol, and version-bound human review. All 617 oracle checks and 75 public-precondition action solves pass. These are software results. Model experiments remain unrun, all 140 visual items await independent human review, and 573 intersection candidates remain unresolved. The contribution is an auditable modular benchmark and measurement repair, with diagnostic localization and physical reliability left as empirical questions.

1. Introduction

An assembly question is meaningful only if both the referenced part and the permitted evidence are explicit. A color judgment from a rendering, a lookup in a connector record, and a count of graph components may refer to the same brick while measuring different operations. A correct graph answer also does not establish a feasible extraction or stability under gravity. Combining such scores without their observation contracts can obscure what a model actually demonstrated.

BrickAtlas uses unchanged original source assemblies (Fig. 1) and source-local numbers such as B0001 to align views, records, graphs and display edits. This supports inspectable comparisons about an exact instance. It does not,

by itself, prove that failures can be localized to independent cognitive modules. Such a claim would require targeted interventions, non-target stability and evidence restoration.

LDraw-2 also addresses a concrete measurement problem in the preceding frozen question generator. In graph deletion and author-step lookup, the correct numerical choice was always second-smallest. Ordered reference arrays additionally encoded answers for type correspondence and neighbors when those arrays were available in the input. Together, these four rules predict 268 of 617 frozen public items. This establishes availability of shortcuts, not that a learned model used them. The old release and any historical evidence remain unchanged.

Our contributions are threefold. First, we preserve an attributable corpus with instance-aligned observation contracts and a complete question-level audit. Second, we repair demonstrated shortcuts and provide a single whitelist boundary with actual request snapshots and independent oracles. Third, we release controlled evidence comparisons, paired rendering and a review workflow that makes remaining ambiguity explicit. The present development manuscript reports measured corpus and software properties; its model and human result cells remain blank.

2. Related Work

Brick assembly and spatial reasoning. BrickNet [3] learns graph-backed generative brick assembly. We reuse its public connector implementation for auditing fixed source placements. Our immediate research question concerns answers and editing contracts on those placements, so our checks do not replicate its generative evaluation. LEGO-Puzzles [6] studies elementary spatial questions and multi-step planning. Its public abstract motivates a related capability question; our narrower contribution is the explicit separation of rendered, source, graph and edit evidence. We do not infer absence of instance identity or diagnostic support in neighboring work.

Physical evaluation. PhyBlock [5] evaluates physical understanding and planning in robotic block assembly.



LDraw-2 currently specifies local graph operations and visibility transitions at fixed poses. Consequently, physical success under PhyBlock’s contract and exact success here are different measured outcomes. Detailed capability comparisons require full protocol alignment and remain to verify.

Diagnostic benchmarks versus capability benchmarks. CLEVR [2] and GQA [1] make compositional question operations explicit. This motivates documenting our operations, but neither their balancing nor their diagnostic validity transfers automatically. A capability score describes performance under an observation contract. Diagnostic localization requires additional evidence that a controlled change affects the intended component while other components remain stable. LDraw-2 supplies intervention infrastructure; this validity study has not been run. Table 1 separates research questions from verified evidence instead of assigning unsupported negative capabilities to other benchmarks.

3. Representation and Task Contracts

3.1. Source Identity and Observation Permissions

An expanded assembly contains instances $(b_i, p_i, c_i, T_i, h_i)$: identity, source part type, color, transform and hierarchy path. A complete reference is the source SHA256 and instance identity; the display label B0001 is unique only within that source. Nested MPD transforms and dependency geometry are retained under the LDraw specification [4]. Independent reparsing verifies type, color, transform and hierarchy for all 15,334 instances. The interactive viewer uses Three.js [7].

V items receive numbered PNG bytes and question/options, with no source attributes. S items receive required coordinates, coverage flags, evidence scope or author-step rows. G items receive a connector record or local edges. E items receive action facts and budget. None

receives reference answers. The human review website additionally exposes internal evidence and is therefore not a model test sandbox.

The recognized connector graph is the undirected projection of matched part pairs. Repeated connector records are deduplicated for graph operations. Missing connector definitions remain unknown evidence, and a scene may contain legitimate separate objects. Intersection candidates without a recognized mating pair are retained for inspection; they are neither confirmed defects nor force estimates.

3.2. Eleven Separate Families

Table 2 states each required input and operation. Color and part-type correspondence are the only two visual families. Their oracle uses source color or part-type equality, which does not certify that a human can distinguish the rendered alternatives. Distance uses supplied bounding-box centers in millimeters and Euclidean arithmetic. Candidates are deterministically hash-sampled; the task does not estimate distances from pixels.

Interface directly reads `connectorRecord.family`; coverage reads a false support flag; author-step lookup finds the first supplied expanded index containing the target. These are evidence controls. Neighbors extracts the exact one-hop candidate set. Graph deletion removes a named vertex from the supplied local graph and counts remaining components, including isolated nodes. It is graph-internal, not a cross-modal task or physical extraction test. The local deletion graphs contain only 5–9 nodes and 2–6 unique edges; their source assemblies can be much larger.

Evidence-limit items all answer “Not established by this evidence” to a gravitational-stability claim. All seventeen source-sequence items also share the same six-step response program. Their scores must be reported as fixed-answer or fixed-program controls. Restoration instead follows the

Table 1. Conservative positioning. External rows describe verified high-level topics. “To verify” means no full capability comparison has been established; it does not mean an absent capability.

Work	Research question	Observation / evaluation contract	Evidence in this study
BrickNet	Generative brick assembly	Connectivity-based build programs	Public connector implementation reused; finer comparison to verify
LEGO-Puzzles	Multi-step spatial reasoning	Visual questions and assembly planning	Public abstract checked; detailed identity/diagnosis support to verify
PhyBlock	Physical understanding and planning	Robotic 3D block assembly	Public abstract checked; fine-grained protocol comparison to verify
CLEVR / GQA	Compositional visual question answering	Images and explicit question operations	Motivation only; no transferred diagnostic validation
LDraw-2	Separate instance-aligned competencies	V / S / G / E, one contract per item	Full software audit; model, human and diagnostic studies unrun

Table 2. All eleven families and their direct measurement contracts. A/M/P/GI denote atomic, metacognitive, procedural and graph-internal organization; they are not calibrated difficulty levels. †: constant semantic response/program; ‡: explicit evidence reading or arithmetic.

Family	N	Layer	Input	Measurement	Direct operation	Necessary input	Shortcut control
Color	73	A	V	Perception	Identify color	Numbered image	Request whitelist
Part type	67	A	V	Perception	Match source type	Numbered image	No reference order
Distance‡	73	A	S	Evidence reading / arithmetic	Euclidean minimum	Centers (mm)	No source path
Interface‡	87	M	G	Evidence reading	Read family	Connector record	Fixed five options
Neighbors	73	M	G	Graph operation	One-hop exact set	Local edges	No reference order
Graph deletion	73	GI	G	Graph operation	Delete; count components	Nodes and local edges	Integer output
Coverage‡	17	M	S	Evidence reading	Read support flag	Coverage rows	Minimal row fields
Evidence limit†	24	M	S	Evidence scope control	Fixed scope response	Available evidence	Report fixed prior
Step lookup†	55	P	S	Evidence reading / lookup	First index lookup	Author-step rows	Legal balanced ranks
Step sequence†	17	P	E	Execution contract	Follow fixed program	Facts and actions	Report fixed prior
Restoration	58	P	E	Execution contract	Restore visibility	Facts and actions	Remove redundancy

public missing/present facts to restore one numbered instance’s visibility.

The four organization layers contain 213 atomic, 201 metacognitive, 130 procedural and 73 graph-internal items. Response formats are 396 single-choice, 73 multiple-choice, 73 integer and 75 action items. These counts reflect applicability, not balanced quotas.

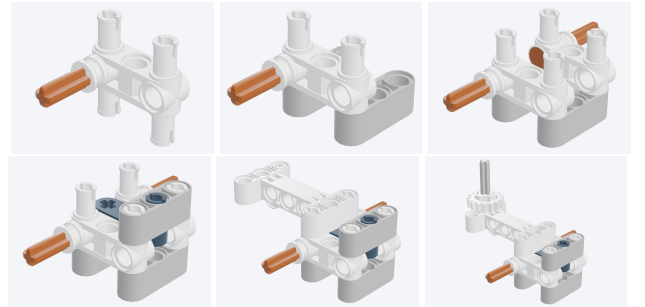
3.3. Executable Scene Edits

For facts F , an action requires $R_a \subseteq F$, forbids $B_a \cap F \neq \emptyset$, and costs c_a within the remaining budget. Accepted execution applies

$$F' = (F \setminus D_a) \cup A_a. \quad (1)$$

The display changes visibility at original source poses. Unknown actions, violated prerequisites, forbidden facts and over-budget execution are rejected without applying that transition. Success requires all goal facts and no prohibited facts.

Figure 2 shows the same six author-window states as the



preserved source replay for set 42102. The v2 facts and semantic program remain unchanged. This is following a supplied author order, not discovering a collision-free insertion sequence. The supplement also shows accepted and rejected restoration at the original poses.

Table 3. Measured size-band by observation counts. Source size and single-item difficulty are separate quantities.

Band	Sources	V	S	G	E	Total
D1	5	17	21	33	12	83
D2	9	34	47	69	25	175
D3	2	14	15	23	7	59
D4	8	75	86	108	31	300

4. Measurement Repair and Corpus Analysis

4.1. Shortcuts and Request Isolation

For author-step lookup, four distinct legal indices now give correct semantic ranks 14/14/14/13. Graph deletion has a boundary constraint: 63 of 73 answers equal the maximum legal component count. Four distinct legal numeric choices would make these answers largest, replacing one shortcut with another. We therefore use exact integer responses for all 73 items, preserving graphs and semantic answers. This is an explicit output-format change, so old numerical-choice scores cannot be silently compared as equivalent conditions.

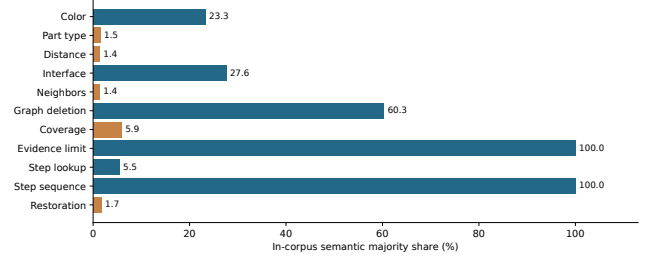
Type-match internal references are globally hash-permuted; correct reference positions are 14/13/13/13/14. References for all families are excluded from actual model requests, eliminating the reference-slice channel. Internal accidental matches remain reported in the audit. Interface now always offers all five connector families; the old answer-dependent option count otherwise exposed stud answers.

The unique serializer admits only a fixed system prompt, English question, format, necessary input, anonymous option IDs/labels and declared image bytes. It removes internal references, target and capability metadata, answer fields, source paths, option values and redundant missing-instance lists. Public preconditions remain because they define the E contract. The runner rejects internal bundle paths, including realpath aliases. Every actual wire request is saved and hashed before an optional API call.

4.2. Source Scale and Task Dependence

The retained corpus has 24 distinct set families and 15,334 instances. Source sizes range from 53 to 1,978, with median 274. D1 is at most 150 instances; D2 is 151–400; D3 is 401–1,000; D4 exceeds 1,000. Table 3 reports the measured cross-tab. Only two sources are in D3. Larger sources generate more items (Pearson $r = 0.9498$, Spearman $\rho = 0.9049$), partly by construction. Neither association measures cognitive difficulty.

Among 593 items with nonempty operands, 148 share their exact source-local operand set with another item. The focal color/type/ distance/restoration families contain 271 items over 73 source-local targets. Shared instances mo-



tivate paired reporting and source-cluster uncertainty; repeated labels across sources are not identical objects. No training split or pretraining-contamination estimate is established.

4.3. Semantic Answer Concentration

Figure 3 shows every family’s in-corpus majority share. Instance-valued responses use source-qualified identities (orange); their low shares are not comparable to a reusable global semantic class and do not establish diversity of reasoning. The graph-deletion majority is 4 on 44/73 items (60.27%). Evidence limit and the six-step program have 100% semantic priors. Across the 396 single-choice items, always A scores 25.76% and uniform sampling has expected accuracy 26.95%. These descriptive controls are not held-out results.

5. Evaluation Protocol and Evidence Status

5.1. Controlled Observations

Standard inputs cover V=140, S=169, G=233 and E=75. Contrary to the prior planning text, the frozen standard visual images already isolate operands. V2 adds full-scene and background-mask images with identical camera, scale, lighting and label projection; the crop separately refits the camera. Figure 4 shows these three observations, left to right, for one type-match item. A crop changes both visibility and scale, so it is not a pure background-effect estimate.

The 14 conditions in Table 4 cover 3,147 eligible item-condition pairs. Edge withholding and ordering apply to 146 edge-based items, excluding 87 interface records. Graph interventions remove a neighbor pair or join surviving components; independent recomputation confirms a changed answer for every eligible item. These are hypothetical evidence graphs, not repaired CAD sources. ID permutation applies to 576 labeled items; the remaining 41 controls contain no B-number. Choice ordering applies to 469 items. Fixed two-view access is executable; an adaptive camera/tool agent is not implemented by that condition.



Table 4. Executable conditions and eligible counts. Every blank exact success cell means not run. Counts are not performance measurements.

Condition	N	Exact success
standard	617	
text-without-image	140	
graph-edges-withheld	146	
full-scene	140	
background-mask	140	
operand-only	140	
choice-order	469	
edge-order	146	
id-permutation	576	
wrong-image	140	
camera-perturbation	140	
color-nuisance	67	
multi-view	140	
graph-intervention	146	

5.2. Scoring, Adapters and Statistical Interpretation

Single choice requires exact `choiceId`; multiple choice requires the exact set without duplicates. Graph deletion requires an exact safe JSON integer in `value`. E requires legal complete execution reaching all goals. Format validity and goal success are reported separately and do not demonstrate mechanical planning. Missing, malformed, refused and timed-out responses remain failures.

Six candidate adapter configurations specify exact model IDs, transport, decoding, prompts, image size, timeout and revision gates. Moving aliases require an actual frozen revision before live use. Both OpenAI-compatible and Anthropic wire formats are implemented. Images are deterministically letterboxed to 1280 by 800 pixels. Each run saves code/config hashes, git revision, actual requests, responses, usage, latency and strict scores. There is one attempt per item, no hidden retry and no automatic JSON repair. Unknown prices or token usage remain null.

Micro accuracy averages item successes; source macro first averages eligible items within each source and then averages eligible sources. The primary paired estimand is the per-family A-minus-B source-macro difference; micro

Table 5. Prospective candidate-family exact success (%). All model result cells are blank. The supplement gives adapter IDs and additional family, resource and review tables.

Candidate family	V	S	G	E	Micro	Macro
SmolVLM						
Qwen3-VL						
Llama 4						
GPT-4.1						
Gemini						
Claude						

is secondary. The protocol specifies 10,000 paired source-cluster bootstrap resamples, seed 20260918 and percentile 95% intervals. Correct-to-wrong and wrong-to-correct transitions must also be reported. Repeated items or views are not independent clusters.

Compare correct, absent and mismatched images; compare full, masked, cropped and text-only evidence. An interval containing zero permits only “no stable gain detected.” It does not establish equivalence, zero contribution or absence of all scale effects. No equivalence margin has been preregistered. Edge withholding tests information dependence; answer-changing interventions together with invariance under edge ordering provide stronger evidence. These comparisons must be evaluated by family before any declared aggregate.

Historical replay requires authentic saved v1 wire requests and receipt provenance. None is presently available for an eligible comparison. We do not reconstruct an intentionally leaky input and present it as a historical run.

5.3. Measured Software Checks and Pending Review

Independent oracles pass for all 617 items, the public-precondition solver succeeds on 75/75 E items, and 5,154 negative controls pass. Path isolation rejects all 24 internal bundles; 617 canaries and request snapshots pass. All 907 new visual-condition images reproduce byte for byte across a second capture of all 140 visual items. There are 617 actual standard wire snapshots and 14 condition canaries, produced without model API calls.

These checks establish software consistency and obser-

vation reproducibility. The required human queue contains every visual item plus 111 other flagged tasks, with reviewer, decision and rationale still empty. A separate queue binds all 573 physical candidates to source hashes and exact part pairs. Human feedback can be exported, reviewed and reimported with version/content validation. Automated checks never create human approvals.

6. Discussion and Limitations

No family currently requires chaining two observation contracts. Instance alignment supports modular comparisons, but diagnostic localization remains unvalidated until targeted failures, non-target stability and evidence restoration are tested. Products or regressions of isolated module scores cannot establish unmeasured integration.

The corpus is small, uneven and potentially familiar from pretraining. Large assemblies do not automatically make operand-isolated items hard. Several families are elementary evidence controls, and the fixed six-step program has no response diversity. Repaired shortcuts do not prove universal absence of non-target cues. Near-identical part variants, lighting-dependent colors and occlusion still require all-item visual adjudication.

OMR provenance is not physical certification. Unsupported definitions, connector tolerances, approximate intersection geometry and the 573 pending pairs prevent claims of complete physical validity. Display replay neither simulates force nor proves insertion feasibility. All original sources and v1 publication assets remain frozen; any future correction must preserve their attribution and version history. The present release is reviewable development infrastructure, with model rankings, human reliability and physical conclusions still open.

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